

Section 6.3:-

Saturday, February 22, 2025 8:17 AM

4. $f(x) = x^{-1}$, $[1, 4]$

$$V = \pi \int_1^4 [x^{-1}]^2 \cdot dx$$

$$V = \pi \int_1^4 x^{-2} \cdot dx$$

$$V = -\pi x^{-1} \Big|_1^4$$

$$V = -\pi (4^{-1} - 1^{-1})$$

$$V = 2.35 \text{ unit}^3$$

9. $f(x) = \frac{2}{x+1}$, $[1, 3]$

$$V = \pi \int_a^b [f(x)]^2 \cdot dx$$

$$V = \pi \int_1^3 \frac{4}{(x+1)^2} \cdot dx$$

$$V = 4\pi \int_1^3 \frac{1}{x^2+2x+1} \cdot dx$$

$$V = 4\pi [\ln(x^2+2x+1)] \Big|_1^3$$

$$V = 4\pi [\ln(9+6+1) - \ln(1+2+1)]$$

$$V = 4\pi [1.386]$$

$$V = 17.42 \text{ unit}^3$$

12. $f(x) = \sqrt{\cos x \sin x}$, $[0, \frac{\pi}{2}]$

$$V = \pi \int_0^{\pi/2} \cos x \cdot \sin x \cdot dx$$

$$\text{let } u = \sin x$$

$$du = \cos x \cdot dx$$

$$V = \int u \cdot du$$

$$V = \frac{u^2}{2}$$

Replacing $u = \sin x$

$$V = \frac{\sin^2 x}{2} \Big|_0^{\pi/2}$$

$$V = \frac{\sin^2(\pi/2)}{2} - \frac{\sin^2(0)}{2}$$

$$V = 1/2 - 0 = 1/2 \text{ unit}^3$$

14. Which of the integrands (i)-(iv) is used to compute the volume obtained by rotating R about $y = 9$ in Figure 11?

(i) $(9 + f(x))^2 - (9 + g(x))^2$

(ii) $(9 + g(x))^2 - (9 + f(x))^2$

(iii) $(9 - f(x))^2 - (9 - g(x))^2$

(iv) $(9 - g(x))^2 - (9 - f(x))^2$

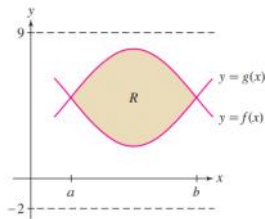


FIGURE 11

• Using washer method:-

$$V = \pi \int_a^b ([\text{outer radius}]^2 - [\text{inner radius}]^2) \cdot dx$$

Outer radius = $g(x) = 9 - g(x)$

inner radius = $f(x) = 9 - f(x)$

$$\text{So, } V = \pi \int_a^b ((9 - g(x))^2 - (9 - f(x))^2) \cdot dx$$

option (iv) is the right option.

51. $y = e^{-x}$, $y = 1 - e^{-x}$, $x = 0$, about $y = 4$

Outer radius = $e^{-x} = a$ Limits = $[0, \infty]$

inner radius = $1 - e^{-x} = b$

as rotating with $y=4$:-

$a = 4 - e^{-x}$, $b = 3 - e^{-x}$

$$V = \pi \int_0^{\infty} ((4 - e^{-x})^2 - (3 - e^{-x})^2) \cdot dx$$

$$V = \pi \int_0^{\infty} [4^2 - 2(4)(e^{-x}) + e^{-2x}] - [9 - 6e^{-x} + e^{-2x}]$$

$$V = \pi \int_0^{\infty} [16 - 8e^{-x} + \cancel{e^{-2x}} - 9 + 6e^{-x} - \cancel{e^{-2x}}] \cdot dx$$

$$V = \pi \int_0^{\infty} [7 - 2e^{-x}] \cdot dx$$

$$V = \pi \int_0^{\infty} [7 - 2e^{-x}] \cdot dx$$

$$V = \int_0^{\infty} 7\pi \cdot dx - \pi \int_0^{\infty} 2e^{-x} \cdot dx$$

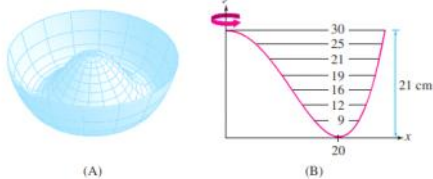
$$V = [7\pi x + \pi 2 e^{-x}]_0^{\infty}$$

$$V = [7\pi(\infty) + \pi 2 e^{-\infty}] - [7\pi(0) + \pi 2]$$

$$V = \infty$$

← confusion

53. The bowl in Figure 14(A) is 21 cm high, obtained by rotating the curve in Figure 14(B) as indicated. Estimate the volume capacity of the bowl shown by taking the average of right- and left-endpoint approximations to the integral with $N = 7$. The inner radii (in centimeters) starting from the top are 0, 4, 7, 8, 10, 13, 14, 20.



$$\text{Thickness} = 21/7 = 3\text{ cm} = \Delta h$$

left:-

$$r = [0, 4, 7, 8, 10, 13, 14]$$

$$V = \pi \sum_{i=0}^{n=7} (y^2) \Delta h$$

$$V = \pi (0^2 + 4^2 + 7^2 + 8^2 + 10^2 + 13^2 + 14^2) \times 3$$

$$V = \pi (16 + 49 + 64 + 100 + 169 + 196) \times 3$$

$$V = \pi (594) \times 3$$

$$V = 1782\text{ cm}^2$$

right-hand approximation:-

$$r = [4, 7, 8, 10, 13, 14, 20]$$

$$V = \pi \sum_{i=1}^{n=8} [y^2] \times \Delta h$$

$$V = \pi [4^2 + 7^2 + 8^2 + 10^2 + 13^2 + 14^2 + 20^2] \cdot 3$$

$$V = [16 + 49 + 64 + 100 + 169 + 196 + 400] \times \pi \times 3$$

$$V = (994) \cdot 3\pi$$

$$V = 9368.22\text{ cm}^3$$

$$V_{avg} = (1782\pi + 2982\pi)/2 = 2382\pi \text{ cm}^3$$

..

56. The **torus** (doughnut-shaped solid) in Figure 15 is obtained by rotating the circle $(x-a)^2 + y^2 = b^2$ around the y -axis (assume that $a > b$). Show that it has volume $2\pi^2 ab^2$. *Hint:* After simplifying it, evaluate the integral by interpreting it as the area of a circle.

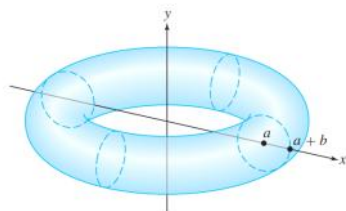


FIGURE 15 Torus obtained by rotating a circle about the y -axis.

as it is revolving around y axis:-

$$(x-a)^2 = b^2 - y^2$$

$$x-a = \pm \sqrt{b^2 - y^2}$$

$$x = a \pm \sqrt{b^2 - y^2}$$

Outer radii:- $x = a + \sqrt{b^2 - y^2} = R$

inner radii:- $x = a - \sqrt{b^2 - y^2} = r$

So, the distance from either side will be,

radius of circle:- $b = r$

$$dV = (\text{area of washers}) dy = \pi (R^2 - r^2) dy$$

$$dV = \pi \left[(a + \sqrt{b^2 - y^2})^2 - (a - \sqrt{b^2 - y^2})^2 \right] dy$$

Section 8.1:-

48. Show that the surface area of a spherical cap of height h and radius R (Figure 19) has surface area $2\pi Rh$.

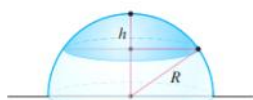


FIGURE 19

from circle:- $x^2 + y^2 = R^2$

$$y = \sqrt{R^2 - x^2}$$

from circle:- $x^2 + y^2 = R^2$

$$y = \sqrt{R^2 - x^2}$$

$$S = 2\pi \int y \sqrt{1 + (y')^2} \cdot dx$$

$$y' = (R^2 - x^2)^{-1/2} = \frac{1}{2(R^2 - x^2)^{3/2}} \cdot (-2x)$$

$$1 + (y')^2 = \frac{R^2}{R^2 - x^2}$$

$$S = 2\pi \int \sqrt{R^2 - x^2} \left(\sqrt{\frac{R^2}{R^2 - x^2}} \right) \cdot dx$$

$$S = 2\pi \int \sqrt{R^2} \cdot dx$$

$$S = 2\pi R x \Big|_{R-h}^R = 2\pi R^2 - 2\pi R^2 + 2\pi R h = \boxed{2\pi R h}$$

Hence proved!

40. $y = e^{-x}$, $[0, 1]$

$$[a, b] = [0, 1]$$

$$\frac{dy}{dx} = -e^{-x}$$

Setting up integral:-

$$\text{Surface area of revolution} = 2\pi \int_a^b f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot dx$$

$$A = 2\pi \int_0^1 e^{-x} \sqrt{1 + (-e^{-x})^2} \cdot dx$$

$$A = 2\pi \int_0^1 e^{-x} \sqrt{1 + e^{-2x}} \cdot dx$$

$$\boxed{A = 4.84 \text{ unit}^2}$$

41. $y = \frac{1}{4}x^2 - \frac{1}{2}\ln x$, $[1, e]$

$$\text{Surface area} = 2\pi \int_a^b f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot dx$$

$$\frac{dy}{dx} = x - \frac{1}{2x}$$

$$du = 2 - 2x$$

$$A = 2\pi \int_1^e \left[\frac{1}{4}x^2 - \frac{1}{2} \ln(x) \right] \sqrt{1 + \left(\frac{x}{2} - \frac{1}{2x} \right)^2} \cdot dx$$

$$\text{Area} = 8.953 \text{ unit}^2$$

49. Find the surface area of the torus obtained by rotating the circle $x^2 + (y - b)^2 = r^2$ about the x -axis (Figure 20).

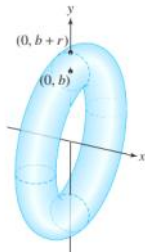


FIGURE 20 Torus obtained by rotating a circle about the x -axis.

$$\text{Surface area} := (2\pi r)(2\pi R)$$

r = smaller radius

R = larger radius.

from the given equation :- $x^2 + (y - b)^2 = r^2$

center $(0, b)$

$$r = r$$

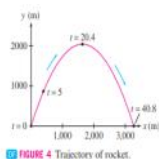
$$R = b$$

So,

$$S = (2\pi r)(2\pi b) = 4\pi^2 rb$$

SECTION 11.1:-

3. Show that the path traced by the bullet in Example 3 is a parabola by eliminating the parameter.



EXAMPLE 3 A model rocket follows the trajectory

$$c(t) = (80t, 200t - 4.9t^2)$$

until it hits the ground, with t in seconds and distance in meters (Figure 4). Find:

(a) The rocket's height at $t = 5$ s.

(b) Its maximum height.

eliminate the parameters:-

$$x = 80t$$

$$y = 200t - 4.9t^2 \quad \text{eq(ii)}$$

$$\left(\frac{x}{80}\right) = t \quad \text{eq(i)}$$

Put this in eq(ii)

$$y = 200\left(\frac{x}{80}\right) - 4.9\left(\frac{x^2}{80^2}\right)$$

$$y = \frac{20}{8}x - \frac{4.9x^2}{6400}$$

$$y = 2.5x - \frac{x^2}{1306.12}$$

$\therefore y = ax + bx^2$, it represents a parabola.

5. Graph the parametric curves. Include arrows indicating the direction of motion.

(a) $(t, t), -\infty < t < \infty$

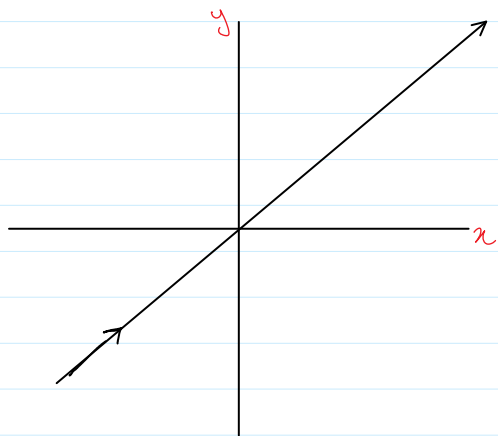
(b) $(\sin t, \sin t), 0 \leq t \leq 2\pi$

(c) $(e^t, e^t), -\infty < t < \infty$

(d) $(t^3, t^3), -1 \leq t \leq 1$

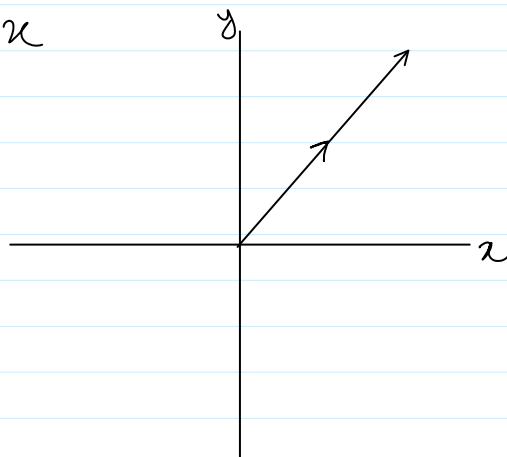
a) $x = t, y = t$

$x = y$ (straight line)



c) $(e^t, e^t) \quad -\infty \leq t \leq \infty$

$y = x$

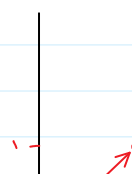


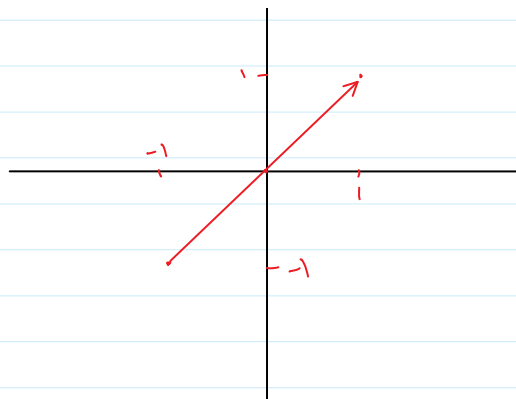
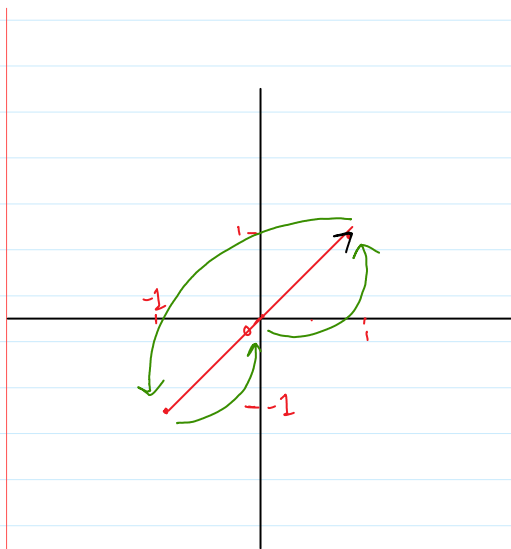
b) $x = \sin t, y = \sin t \quad 0 \leq t \leq 2\pi$

$x = y$

d) $(t^3, t^3) \quad -1 \leq t \leq 1$

$y = x$





7. $x = t + 3, \quad y = 4t$

$$x = t + 3$$

$$t = x - 3$$

put in eq (ii) :-

$$y = 4(x - 3) = 4x - 12$$

19. Match the parametrizations (a)–(d) below with their plots in Figure 15, and draw an arrow indicating the direction of motion.

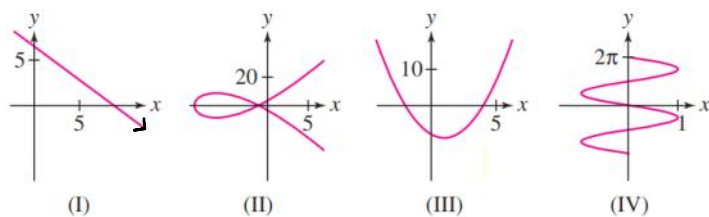


FIGURE 15

(a) $c(t) = (\sin t, -t)$

(b) $c(t) = (t^2 - 9, 8t - t^3)$

(c) $c(t) = (1 - t, t^2 - 9)$

(d) $c(t) = (4t + 2, 5 - 3t)$

$$\begin{aligned} \text{a) } x &= \sin t \\ y &= -t \\ \text{or } y &= -\sin^{-1}(x) \\ x &= -\sin(y) \\ \text{for } a &= IV \end{aligned}$$

$$\begin{aligned} \text{b) } x &= t^2 - 9 \\ y &= 8t - t^3 \\ t &= \sqrt{x+9} \\ y &= 8\sqrt{x+9} - (\sqrt{x+9})^3 \\ b &= II \end{aligned}$$

$$\begin{aligned} \text{c) } x &= 1-t \\ y &= t^2 - 9 \\ t &= 1-x \\ y &= (1-x)^2 - 9 \\ \text{parabola:- } c &= III \end{aligned}$$

$$\begin{aligned} \text{d) } x &= 4t + 2 \\ y &= 5 - 3t \\ t &= \frac{x-2}{4} \\ y &= 5 - 3\left(\frac{x-2}{4}\right) \\ y &= \frac{20 - 3x + 6}{4} \\ 4y &= 26 - 3x \\ d &= I \end{aligned}$$

21. A particle follows the trajectory

$$x(t) = \frac{1}{4}t^3 + 2t, \quad y(t) = 20t - t^2$$

with t in seconds and distance in centimeters.

(a) What is the particle's maximum height?

(b) When does the particle hit the ground and how far from the origin does it land?

To find max height;

$$\begin{aligned} \frac{dy}{dt} &= 20 - 2t \\ &= 20 - 2t \end{aligned}$$

at Max height slope = 0
hence,

$$\begin{aligned} 0 &= 20 - 2t \\ -20 &= -2t \\ t &= 10 \text{ sec.} \end{aligned}$$

Put in equation:- $y = 20t - t^2$

$$y = 20(10) - 10^2$$

$$y = 200 - 100 = 100 \text{ cm}$$

max height = 100 cm.

b) Particle hit the ground when $y(t) = 0$:-

$$0 = 20t - t^2$$

$$t - 20t = 0$$

$$t = 0, t = 20$$

• Particle hit the ground after 20 seconds.

• How far? $x(t) = \frac{1}{4} t^3 + 2t$

$$x(20) = \frac{1}{4} (20)^3 + 2(20) = 2040 \text{ cm}$$

31. Line through (3, 1) and (-5, 4)

$$x_1 = 3$$

$$x_2 = -5$$

$$y_1 = 1$$

$$y_2 = 4$$

$$(y - y_1) = m(x - x_1)$$

$$(y - 1) = \frac{y_2 - y_1}{x_2 - x_1} (x - 3)$$

$$(y - 1) = \frac{-3}{8} (x - 3)$$

$$y = \frac{-3}{8} x + 17/8$$

33. Segment joining (1, 1) and (2, 3)

$$m = y_2 - y_1$$

$$= 2$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 1}{2 - 1} = 2$$

$$\therefore (y - y_1) = m(x - x_1)$$

$$(y - 1) = 2(x - 1)$$

$$y = 2x - 1$$

45. The graphs of $x(t)$ and $y(t)$ as functions of t are shown in Figure 16(A). Which of (I)–(III) is the plot of $c(t) = (x(t), y(t))$? Explain.

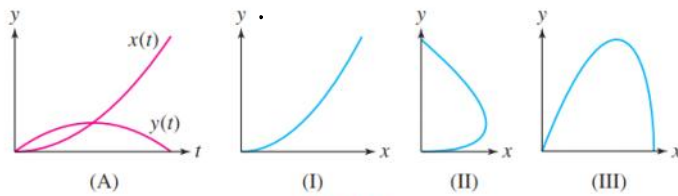


FIGURE 16

• Graph III is the plot of $c(t) = (x(t), y(t))$

- in (A); The graph of $x(t)$ is increasing, while graph of $y(t)$ reaches a peak and starts decreasing. hence plot graph of $c(t)$ should not go backwards in x direction so II is wrong, while first one continuously increases in y -direction which is also wrong. while III makes all of the conditions.

67. Show that the Bézier curve with control points

$$P_0 = (1, 4), \quad P_1 = (3, 12), \quad P_2 = (6, 15), \quad P_3 = (7, 4)$$

has parametrization

$$c(t) = (1 + 6t + 3t^2 - 3t^3, 4 + 24t - 15t^2 - 9t^3)$$

$$B_0^3 = (1-t)^3, \quad B_1^3(t) = 3(1-t)^2t, \quad B_2^3(t) = 3(1-t)t^2, \quad B_3^3 = 3t^3$$

for x :-

$$x(t) = 1(1-t)^3 + 3(3(1-t)^2t) + 6(3(1-t)t^2) + 7t^3$$

$$x(t) = (1 - 3t + 3t^2 - t^3) + (9t - 9t^2 + 3t^3) + (18t^2 - 18t^3) + 7t^3$$

$$x(t) = 1 + 6t + 3t^2 - 3t^3$$

for y:-

$$y(t) = 4(1-t)^3 + 12(3(1-t)^2t) + 15(3(1-t)t^2) + 4t^3$$

$$y(t) = 4 - 12(t) + 12t^2 - 4t^3 + 36t - 36t^2 + 12t^3 + 45t^2 - 45t^3 + 4t^3$$

$$y(t) = 4 + 24t - 15t^2 - 9t^3$$

So,

$$c(t) = (1 + 6t + 3t^2 - 3t^3, 4 + 24t - 15t^2 - 9t^3)$$

76. Property of the Cycloid Prove that the tangent line at a point P on the cycloid always passes through the top point on the rolling circle as indicated in Figure 21. Assume the generating circle of the cycloid has radius 1.

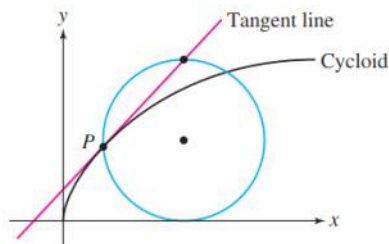


FIGURE 21

Parametric equation of cycloid:-

$$x = t - \sin t \quad y = 1 - \cos t \quad \text{for } r=1$$

$$\frac{dx}{dt} = 1 - \cos t \quad \frac{dy}{dt} = \sin t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \sin t \times \frac{1}{1 - \cos t}$$

$$\frac{dy}{dx} = \frac{\sin(t)}{1 - \cos(t)} \quad \text{--- eq(ii)}$$

$$\therefore \sin(t) = 2 \sin(t/2) \cos(t/2)$$

$$\therefore 1 - \cos(t) = 2 \sin^2(t/2)$$

put both the identities in eq (i) :-

$$\frac{dy}{dx} = \frac{\cancel{2 \sin t/2} \cos(t/2)}{\cancel{2 \sin^2(t/2)}} = \frac{\cos(t/2)}{\sin(t/2)} = \cot(t/2)$$

• equation of tangent line:-

$$y - y_1 = m(x - x_1)$$

$$y - (1 - \cos t) = \cot(t/2)(x - (t - \sin t))$$

• The top of the circle $(x_r, y_r) = (t, 2)$

$$2 - (1 - \cos t) = \cot(t/2)(t - (t - \sin t))$$

$$1 + \cos t = \cot(t/2) \sin t$$

$$1 + \cos(t) = \frac{\cos(t/2)}{\sin(t/2)} \cdot \sin(t)$$

$$1 + \cos(t) = \frac{\cos(t/2) \cdot \cancel{2 \sin(t/2)} \cdot \cos(t/2)}{\cancel{\sin(t/2)}}$$

$$1 + \cos(t) = 2 \cos^2(t/2)$$

- as the equation hold true, hence it will always pass the top point of the rolling circle.

Section 11.2

4. $(3t, 4t^{3/2}), 0 \leq t \leq 1$

$$x = 3t \quad 0 \leq x \leq 3$$

$$t = x/3$$

$$\frac{dx}{dt} = 3 \quad \frac{dy}{dt} = 6\sqrt{t}$$

$$L = \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \cdot dt$$

$$L = \int_0^1 \sqrt{9 + 36t} \cdot dt$$

$$L = 3 \int_0^1 \sqrt{1 + 4t} \cdot dt$$

$$\text{Let } u = 1 + 4t$$

$$du = 4 dt$$

$$L = \frac{3}{4} \int_0^1 \sqrt{u} \cdot du$$

$$L = \frac{3}{4} \left. \frac{u^{3/2}}{3/2} \right|_0^1$$

$$L = \frac{3}{4} \times \frac{2}{3} \cdot (1+4t)^{3/2} \Big|_0^1$$

$$L = \frac{1}{2} \left[(1+4(1))^{3/2} - (1+0)^{3/2} \right]$$

$$L = \frac{1}{2} [11 - 1]$$

$$L = 5.05 \text{ unit.}$$

8. $(\sin \theta - \theta \cos \theta, \cos \theta + \theta \sin \theta), \quad 0 \leq \theta \leq 2$

$$x = \sin \theta - \theta \cos \theta$$

$$y = \cos \theta + \theta \sin \theta \quad 0 \leq \theta \leq 2$$

$$\frac{dx}{d\theta} = \cos\theta - (\cos\theta - \theta \sin\theta) = \theta \sin\theta$$

$$\frac{dy}{d\theta} = -\sin\theta + \sin\theta + \theta \cos\theta = \theta \cos\theta$$

$$\text{Arc length} = \int_0^2 \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} \cdot d\theta$$

$$L = \int_0^2 \sqrt{(\theta \sin\theta)^2 + (\theta \cos\theta)^2} \cdot d\theta$$

$$L = \int_0^2 \sqrt{\theta^2 (\sin^2\theta + \cos^2\theta)} \cdot d\theta$$

$$L = \int_0^2 \cancel{\theta} \cdot d\theta$$

$$L = \left. \frac{\theta^2}{2} \right|_0^2$$

$$L = \left(\frac{4}{2} - \frac{0}{2} \right) = \boxed{2}$$

12. Find the length of the spiral $c(t) = (t \cos t, t \sin t)$ for $0 \leq t \leq 2\pi$ to three decimal places (Figure 7). *Hint:* Use the formula

$$\int \sqrt{1+t^2} dt = \frac{1}{2}t\sqrt{1+t^2} + \frac{1}{2}\ln(t + \sqrt{1+t^2})$$

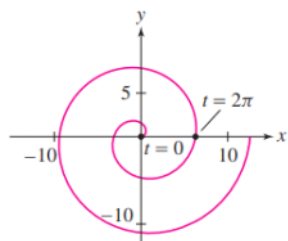


FIGURE 7 The spiral $c(t) = (t \cos t, t \sin t)$.

$$x(t) = t \cos t \quad y(t) = t \sin t$$

$$\frac{dx}{dt} = -t \sin t + \cos t \quad \frac{dy}{dt} = \sin t + t \cos t$$

$\frac{d}{dt}$ dt

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \cdot dt$$

$$L = \int_0^{2\pi} \sqrt{(-t \sin t + \cos t)^2 + (\sin t + t \cos t)^2} \cdot dt$$

$$L = \int_0^{2\pi} \sqrt{t^2 \sin^2 t - 2t \sin t \cos t + \cos^2 t + \sin^2 t + 2t \sin t \cos t + t^2 \cos^2 t} \cdot dt$$

$$L = \int_0^{2\pi} \sqrt{t^2 \sin^2 t + t^2 \cos^2 t + \sin^2 t + \cos^2 t} \cdot dt$$

$$L = \int_0^{2\pi} \sqrt{t^2 (\sin^2 t + \cos^2 t) + (\sin^2 t + \cos^2 t)} \cdot dt$$

$$\therefore \sin^2 t + \cos^2 t = 1$$

$$L = \int_0^{2\pi} \sqrt{t^2 + 1}$$

from question:-

$$L = \left[\frac{1}{2} t \sqrt{1+t^2} + \frac{1}{2} \ln(t + \sqrt{1+t^2}) \right]_0^{2\pi}$$

$$L = \left[\frac{1}{2} (2\pi) \sqrt{1+(2\pi)^2} + \frac{1}{2} \ln(2\pi + \sqrt{1+(2\pi)^2}) \right] - \left[\frac{1}{2} (0) + \frac{1}{2} \ln(0 + \sqrt{1}) \right]$$

$$L = 19.987 + 1.268 = 21.255$$

24. Calculate the arc length integral $s(t)$ for the logarithmic spiral $c(t) = (e^t \cos t, e^t \sin t)$.

$$x = e^t \cos t$$

$$y = e^t \sin t$$

$$\frac{dx}{dt} = e^t \cos t - e^t \sin t$$

$$\frac{dy}{dt} = e^t \sin t + e^t \cos t$$

$$S(t) = \int_a^b \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} \cdot dt$$

$$S(t) = \int_a^b \sqrt{(e^t \sin t + e^t \cos t)^2 + (e^t \cos t - e^t \sin t)^2} \cdot dt$$

$$S(t) = \int_a^b \sqrt{e^{2t} \sin^2 t + 2e^{2t} \sin t \cos t + e^{2t} \cos^2 t + e^{2t} \cos^2 t - 2e^{2t} \sin t \cos t + e^{2t} \sin^2 t}$$

$$S(t) = \int_a^b \sqrt{2e^{2t} \sin^2 t + 2e^{2t} \cos^2 t} \cdot dt$$

$$S(t) = \int_a^b \sqrt{2e^{2t} (\sin^2 t + \cos^2 t)} \cdot dt$$

$$S(t) = \int_a^b \sqrt{2e^{2t}} \cdot dt$$

$$S(t) = \sqrt{2} \int_a^b e^t \cdot dt$$

Section 11.4

4. Find the area of the shaded triangle in Figure 13 as an integral in polar coordinates. Then find the rectangular coordinates of P and Q , and compute the area via geometry.

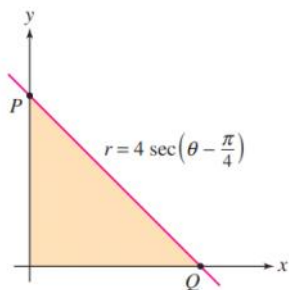


FIGURE 13

Area in polar coordinate system:- (by integrals)

$$A = \int_{\theta_1}^{\theta_2} [r(\theta)]^2 \cdot d\theta$$

• as triangle's range is from 0 to $\pi/2$:-

$$A = \frac{1}{2} \int_0^{\pi/2} [4 \sec(\theta - \pi/4)]^2 \cdot d\theta$$

$$A = \frac{1}{2} \times 16 \int_0^{\pi/2} \sec^2(\theta - \pi/4) \cdot d\theta$$

$$A = 8 \left[\tan(\theta - \pi/4) \right] \Big|_0^{\pi/2}$$

$$A = 8 \left(\tan(\pi/2 - \pi/4) - \tan(0 - \pi/4) \right)$$

$$A = 8 (1 + 1) = 8 \times 2 = 16 \text{ unit}^2$$

• Area by geometry:-

$$A = \frac{1}{2} b \times h$$

• finding rectangular coordinates of P, Q:-

• for $(\theta=0)$:-

$$r = 4 \sec(\theta - \pi/4) = 4 \times \frac{1}{\cos(\theta - \pi/4)} = 4\sqrt{2}$$

$$\cos(-\pi/4)$$

So, $Q = (4\sqrt{2}, 0)$

• for $(Q = \pi/2)$:-

$$r = 4 \sec\left(\pi/2 - \pi/4\right) = 4 \sec \frac{\pi}{4} = 4\sqrt{2}$$

So, $P = (0, 4\sqrt{2})$

distance along x -axis = base = $4\sqrt{2}$
distance along y -axis = height = $4\sqrt{2}$

$$A = \frac{1}{2} \times 4\sqrt{2} \times 4\sqrt{2} = 4 \times 4 = \boxed{16} \text{ unit}^2$$

6. Which interval of θ -values corresponds to the shaded region in Figure 15? Find the area of the region.

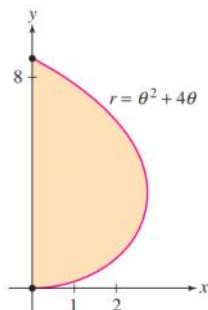


FIGURE 14

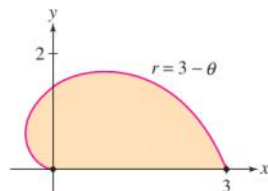


FIGURE 15

• To find the interval of θ :-

$$r = 3 - \theta$$

as ' r ' must be a non-negative number :-

$$3 - \theta \geq 0$$

$$3 \geq \theta$$

as ' θ ' starts from positive x, y -axis;
hence the smallest value could be '0'

$$0 \leq \theta \leq 3$$

• for area :-

- for area:-

$$A = \frac{1}{2} \int_0^3 [r(\theta)]^2 \cdot d\theta$$

$$A = \frac{1}{2} \int_0^3 (3-\theta)^2 \cdot d\theta$$

$$A = \frac{1}{2} \int_0^3 (9 - 6\theta + \theta^2) \cdot d\theta$$

$$A = \frac{1}{2} \left[\int_0^3 9 \cdot d\theta - 6 \int_0^3 \theta \cdot d\theta + \int_0^3 \theta^2 \cdot d\theta \right]$$

$$A = \frac{1}{2} \left[9\theta \Big|_0^3 - 6 \frac{\theta^2}{2} \Big|_0^3 + \frac{\theta^3}{3} \Big|_0^3 \right]$$

$$A = \frac{1}{2} \left[9(3-0) - 3(3^2 - 0^2) + \frac{(3^3) - (0)^3}{3} \right]$$

$$A = \frac{1}{2} [27 - 27 + 9] = 4.5 \text{ unit}^2$$

7. Find the total area enclosed by the cardioid in Figure 16.

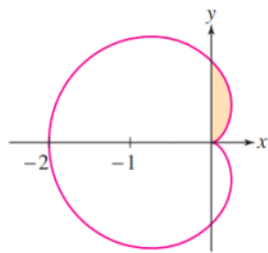


FIGURE 16 The cardioid $r = 1 - \cos \theta$.

- Total area enclosed by cardioid:-

$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} [r]^2 \cdot d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (1 - \cos \theta)^2 \cdot d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (1^2 - 2 \cos \theta + \cos^2 \theta) \cdot d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (1^2 - 2 \cos \theta + \cos^2 \theta) \cdot d\theta$$

$$A = \frac{1}{2} \left[\int_0^{2\pi} 1 \cdot d\theta - 2 \int_0^{2\pi} \cos \theta \cdot d\theta + \int_0^{2\pi} \cos^2 \theta \cdot d\theta \right]$$

$$A = \frac{1}{2} \left[\theta \Big|_0^{2\pi} + 2 \sin(\theta) \Big|_0^{2\pi} + \int_0^{2\pi} \left[\frac{1}{2} + \frac{\cos 2\theta}{2} \right] \cdot d\theta \right]$$

$$A = \frac{1}{2} \left[(2\pi) + 0 + \pi \right]$$

$$A = \frac{3\pi}{2} \text{ unit}^2$$

13. Find the area of region A in Figure 19.

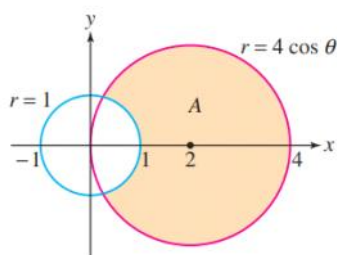


FIGURE 19

first we have to find where both circles meet:-

$$4 \cos \theta = 1$$

$$\cos \theta = 1/4$$

$$\theta = \cos^{-1}(1/4)$$

$$\text{limits of } -\cos^{-1}(1/4) \leq \theta \leq \cos^{-1}(1/4)$$

$$\text{Area} = \frac{1}{2} \int_{-\cos^{-1}(1/4)}^{\cos^{-1}(1/4)} \left[(4 \cos \theta)^2 - (1)^2 \right] \cdot d\theta$$

$$\text{Area} = \frac{1}{2} \left[\frac{16}{2} \sin 2\theta - \theta \right]_{-\cos^{-1}(1/4)}^{\cos^{-1}(1/4)}$$

$$\text{Area} = \frac{1}{2} \int_{-\cos^{-1}(1/4)}^{\cos^{-1}(1/4)} [16 \cos^2 \theta - 1] \cdot d\theta$$

$$\text{Area} = 11.16 \text{ unit}^2$$

14. Find the area of the shaded region in Figure 20 enclosed by the circle $r = \frac{1}{2}$ and a petal of the curve $r = \cos 3\theta$. *Hint:* Compute the area of both the petal and the region inside the petal and outside the circle.

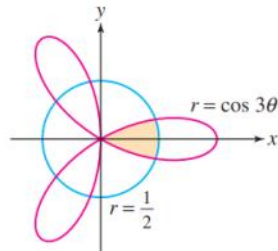


FIGURE 20

Total area of petal:-

each petal starts at origin and ends at origin so,

$$\cos(3\theta) = 0$$

$$3\theta = \cos^{-1}(0) = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

for one petal :- $-\frac{\pi}{6} \leq \theta \leq \frac{\pi}{6}$

$$\text{Area} = \frac{1}{2} \int_{-\pi/6}^{\pi/6} (\cos 3\theta)^2 \cdot d\theta$$

$$\text{Area} = \frac{1}{2} \int_{-\pi/6}^{\pi/6} \frac{1 + \cos 6\theta}{2} \cdot d\theta$$

$$\text{Area} = \frac{1}{2} \int_{-\pi/6}^{\pi/6} (\cos 3\theta)^2 \cdot d\theta$$

$$\text{Area} = \frac{1}{2} \int_{-\pi/6}^{\pi/6} \frac{1 + \cos 6\theta}{2} \cdot d\theta$$

$$A = \frac{1}{2} \left[\int_{-\pi/6}^{\pi/6} \frac{1}{2} \cdot d\theta + \int_{-\pi/6}^{\pi/6} \frac{\cos 6\theta}{2} \cdot d\theta \right]$$

$$A = \frac{1}{2} \left[\frac{\theta}{2} \Big|_{-\pi/6}^{\pi/6} + \frac{\cos 6\theta}{12} \Big|_{-\pi/6}^{\pi/6} \right]$$

$$A = \frac{1}{2} \left[\frac{2\pi}{12} \right] = \frac{\pi}{12} \text{ unit}^2$$

• Area for circle:-

$$A = \int_{-\pi/6}^{\pi/6} \left(\frac{1}{2} \right)^2 \cdot d\theta$$

$$A = \frac{1}{8} \int_{-\pi/6}^{\pi/6} d\theta$$

$$A = \frac{1}{8} \cdot \frac{2\pi}{6} = \frac{\pi}{24} \text{ unit}^2$$

• Shaded area = $A_{\text{petal}} - A_{\text{circle}} = \frac{\pi}{12} - \frac{\pi}{24} = \boxed{\frac{\pi}{24} \text{ unit}^2}$

22. Find the area of the region that lies inside one but not both of the curves in Figure 23.

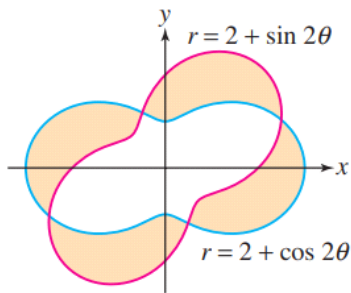


FIGURE 23

Area of first curve:- $A = \frac{1}{2} \int_0^\pi (2 + \sin(2\theta))^2 \cdot d\theta$

$$\text{Area} = \frac{1}{2} \int_0^\pi (4 + 4\sin 2\theta + \sin^2(2\theta)) d\theta$$

$$\text{Area} = \frac{1}{2} \int_0^\pi \left(4 + 4\sin 2\theta + \frac{1 - \cos(4\theta)}{2} \right) \cdot d\theta$$

$$A = \frac{1}{2} \int_0^\pi 9 + \frac{8\sin 2\theta}{2} - \frac{\cos 4\theta}{2} \cdot d\theta$$

$$A = \frac{1}{4} \left[\int_0^\pi 9 \cdot d\theta + 8 \int_0^\pi \sin 2\theta \cdot d\theta - \int_0^\pi \cos 4\theta \cdot d\theta \right]$$

$$A = \frac{1}{4} \left[9\theta \Big|_0^\pi + 8 \frac{\sin 2\theta}{2} \Big|_0^\pi - \frac{\cos 4\theta}{4} \Big|_0^\pi \right]$$

$$A = \frac{1}{4} \left[9(\pi) + 8 \left[\frac{\sin(2\pi)}{2} - \frac{\sin(0)}{2} \right] - \frac{\cos(4\pi)}{4} - \frac{\cos(0)}{4} \right]$$

$$A = \frac{1}{4} \left[9\pi + 8[\cancel{0}] - [\cancel{\frac{2}{4}} - \frac{1}{4}] \right]$$

$$A = \frac{9\pi}{4} = 7.06$$

for whole petal:- $A \times 2 = 7.06 \times 2 = 14.13 \text{ unit}^2$

as both petals are similar, their Area will be similar

Area of $r_1, r_2 = 14.13 \text{ unit}^2$

Now,
find the area under both the curves:-

finding the minimum function.

$$2 + \sin 2\theta = 2 + \cos 2\theta$$

$$\frac{\sin 2\theta}{\cos 2\theta} = 1$$

$$\tan 2\theta = 1$$

$$2\theta = \tan^{-1}(1)$$

$$2\theta = \pi/4$$

$$\theta_1 = \pi/6$$

$$\theta_2 = 2\pi/3$$

area under $0 \leq \theta \leq \pi$:-

$$A = \frac{1}{2} \int_0^{\pi/6} (2 + \sin 2\theta)^2 \cdot d\theta + \frac{1}{2} \int_{\pi/6}^{2\pi/3} (2 + \cos 2\theta)^2 \cdot d\theta + \frac{1}{2} \int_{2\pi/3}^{\pi} (2 + \sin 2\theta) \cdot d\theta$$

$$A = 1.623 + 0.91 + 1.802$$

$$A = 4.33 \times 2 = 8.67$$

$$\text{shaded area} = \text{Total} - \text{petal}$$

$$= 14.13 + 14.13 - 8.67$$

$$= 19.59 \text{ unit}^2$$

30. The cardioid $r = 1 - \cos \theta$ in Figure 16

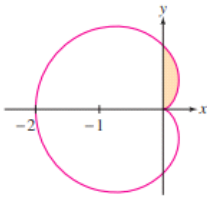


FIGURE 16 The cardioid $r = 1 - \cos \theta$.

$$L = \int_a^b \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2} \cdot d\theta$$

$$\frac{dr}{d\theta} = \sin \theta$$

$$L = \int_0^{2\pi} \sqrt{(\sin \theta)^2 + (1 - \cos \theta)^2} \cdot d\theta$$

$$L = \int_0^{2\pi} \sqrt{\cancel{\sin^2 \theta} + 1 - 2\cos \theta + \cancel{\cos^2 \theta}} \cdot d\theta$$

$$L = \int_0^{2\pi} \sqrt{2 - 2\cos \theta} \cdot d\theta$$

$$L = \int_0^{2\pi} \sqrt{2(1 - \cos \theta)} \cdot d\theta$$

$$\therefore 2(1 - \cos \theta) = 4 \sin^2(\theta/2)$$

$$L = \int_0^{2\pi} \sqrt{4 \sin^2(\theta/2)} \cdot d\theta$$

$$L = \int_0^{2\pi} 2 \sin(\theta/2) \cdot d\theta$$

$$L = \int_0^{\pi} 4 \sin(u) \cdot du$$

$$L = -4 \cos u \Big|_0^{\pi}$$

$$L = -4(-1-1) = -4(-2) = 8$$

$$L = 8$$

In Exercises 33–36, express the length of the curve as an integral but do not evaluate it.

33. $r = e^\theta + 1, \quad 0 \leq \theta \leq \pi/2.$

$$\therefore L = \int_a^b \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2} \cdot d\theta$$

$$L = \int_0^{\pi/2} \sqrt{e^{2\theta} + (e^\theta + 1)^2} \cdot d\theta$$

36. $r = \sin \theta \cos \theta, \quad 0 \leq \theta \leq \pi$

$$L = \int_0^\pi \sqrt{\cos^2 2\theta + \sin^2 \theta} \cdot \cos^2 \theta \cdot d\theta$$